

PRACTICE TEST

Similar Problems

Pseudocode exercises, modeled after Computer Science Baccalaureate exam problems

Problem 1 (10p)

Write, in pseudocode, a program sequence that reads two integers x (with exactly 5 digits) and y (with exactly 4 digits) from the keyboard, and displays the message YES if the number formed, in this order, from the 4th digit of x , the 1st digit of y , and the 3rd digit of x , is divisible by 9, or the message NO otherwise. Digits of a number are numbered from right to left, starting with 1 for the units digit.

Example: if $x = 73621$ and $y = 4809$, the answer is YES (the formed number, 396, is divisible by 9).

Problem 2 (10p)

For the requirement below, identify the input data, the output data, and write the associated program sequence.

Consider six natural numbers $h1, m1, s1$ and $h2, m2, s2$, representing the duration of two time intervals, expressed respectively in hours, minutes, and seconds (with $0 \leq m1, m2, s1, s2 \leq 59$). Read the values $h1, m1, s1$ and $h2, m2, s2$, with the meaning above, and display, in order, the hours, minutes, and seconds of the sum of the two intervals, separated by single spaces.

Example: if the values read are 1 45 50 and 2 30 15, the screen displays: 4 16 5

Problem 3 (15p)

In the program sequence below, all variables are integers. A sequence of 10 natural values, each with at most 4 digits, is entered from the keyboard; the sequence contains at least one multiple of 3 and at least one number that is not a multiple of 3. Write an expression that can replace the dotted area so that, after the resulting sequence is executed, the value of the largest multiple of 3 read is displayed.

```
y ← -1;
for i ← 10, 1, -1 do
{ read x;
  if (.....) then
    y ← x
}
write y;
```

Problem 4 (15p)

Maria wants to read a book with n pages. On the first day she reads 2 pages, and on each following day she reads 2 pages more than the previous day (that is, 2, 4, 6, 8, ...). Find the number of days it takes Maria to finish reading the book.

Task: Write a program sequence that reads a natural number n from the keyboard and displays the number of days it takes Maria to finish the book.

Constraints: $10 \leq n \leq 10000$. On the last day, Maria may read fewer pages than calculated, if she already finishes the book by then.

Solutions

The solutions below use the standard pseudocode structures: read, write, assignment ($\leftarrow$), if...then...else, for, while, do...while, and repeat...until.

Solution — Problem 1

```
read x, y
d1  $\leftarrow$  (x / 1000) % 10
d2  $\leftarrow$  y % 10
d3  $\leftarrow$  (x / 100) % 10
number  $\leftarrow$  d1*100 + d2*10 + d3
if number % 9 = 0 then
    write 'YES'
else
    write 'NO'
```

Solution — Problem 2

Input data: $h1, m1, s1, h2, m2, s2$ — natural numbers, with $0 \leq m1, m2, s1, s2 \leq 59$.

Output data: H, M, S — the hours, minutes, and seconds of the sum of the two intervals.

```
read h1, m1, s1, h2, m2, s2
totalS  $\leftarrow$  s1 + s2
S  $\leftarrow$  totalS % 60
totalM  $\leftarrow$  m1 + m2 + totalS / 60
M  $\leftarrow$  totalM % 60
H  $\leftarrow$  h1 + h2 + totalM / 60
write H, ' ', M, ' ', S
```

Solution — Problem 3

```
x % 3 = 0 AND x > y
```

Solution — Problem 4

```
read n
total  $\leftarrow$  0
pages  $\leftarrow$  0
day  $\leftarrow$  0
do
{
    pages  $\leftarrow$  pages + 2
    total  $\leftarrow$  total + pages
    day  $\leftarrow$  day + 1
}
while total < n
write day
```